

Production of mono and bilayer graphene

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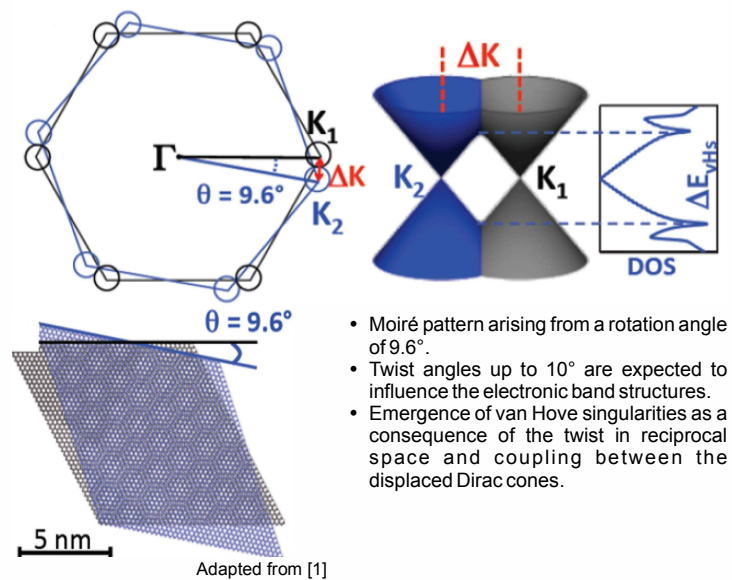
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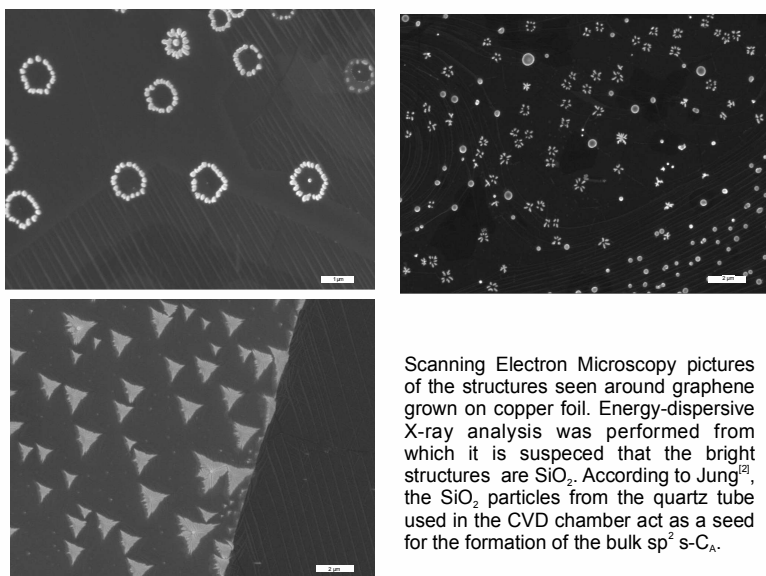
Abstract

Twisted bilayer graphene is a material of great interest due to its electronic structure varying with small twist angles. Selective growth of single crystalline areas is one of the main challenges in graphene research. This contribution reports reliable growth of mono and twisted bi-layered graphene structures based on chemical vapour deposition (CVD). The quality of layered graphene grown by decomposition of methane on copper can be altered by changing the flow rates of gases, degree of dilution of the precursor methane, pretreatment of catalyst surface, temperature and time of growth. Annealing conditions play an important role in providing the nucleating sites for growth of hexagonal graphene. Combination of methane flow rate and time governs the dimensions of the grown layers.

Twisted Bilayer Graphene

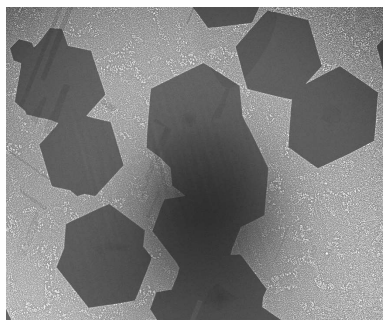


Nanostructures during CVD



Transfer of CVD grown graphene

- Polymers like PMMA (poly-methylmethacrylate)^[3], polystyrene, polycarbonate are used to transfer CVD grown graphene from copper foils to desired substrates.

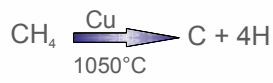


After growth on copper foil.

After transfer on Si substrate.

Chemical Vapor Deposition

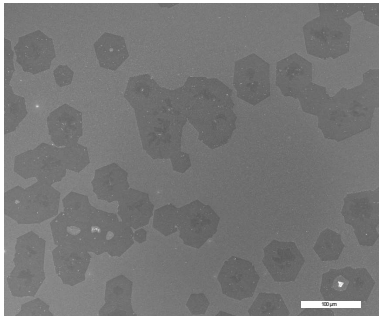
Chemical reaction:



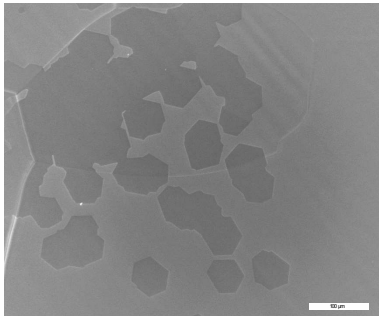
Steps:

- Cleaning of copper foil
- Annealing of the copper foil
- Growth of graphene by CVD

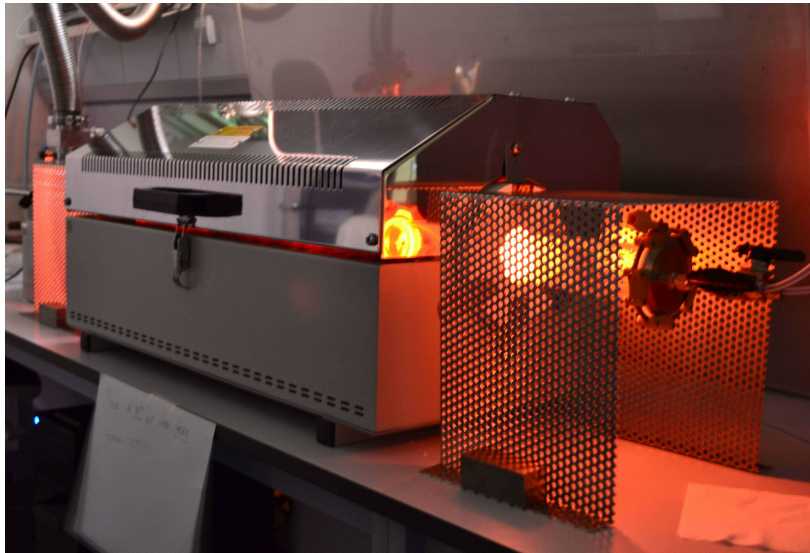
Oxygen deficient copper



Oxygen rich copper

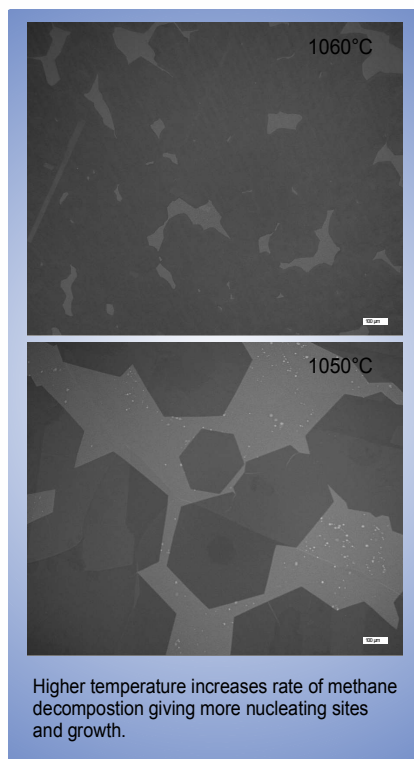
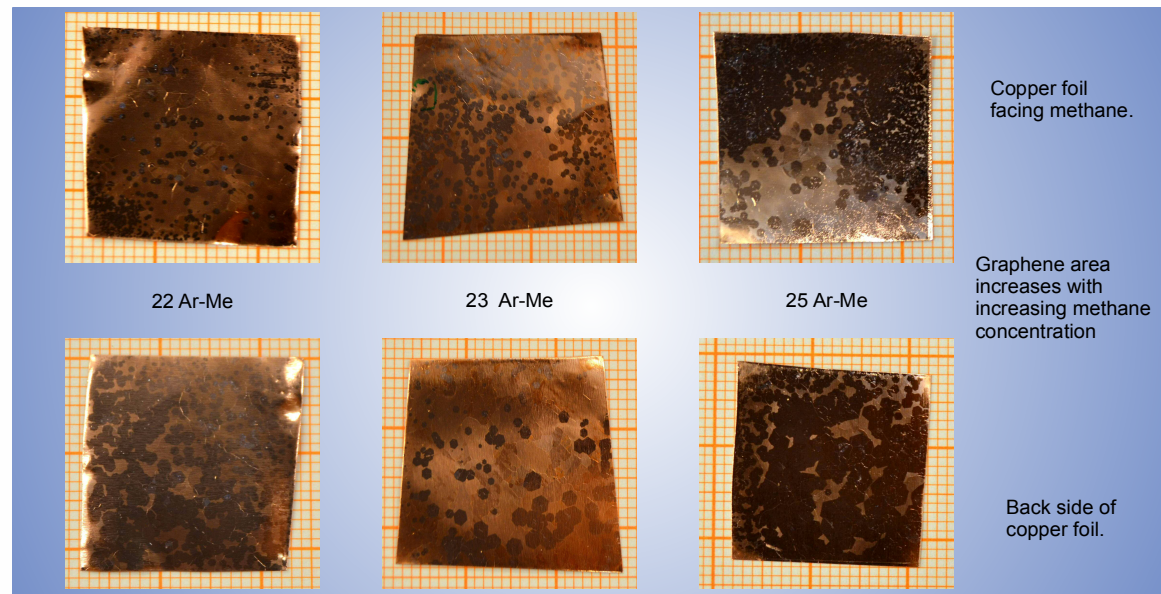
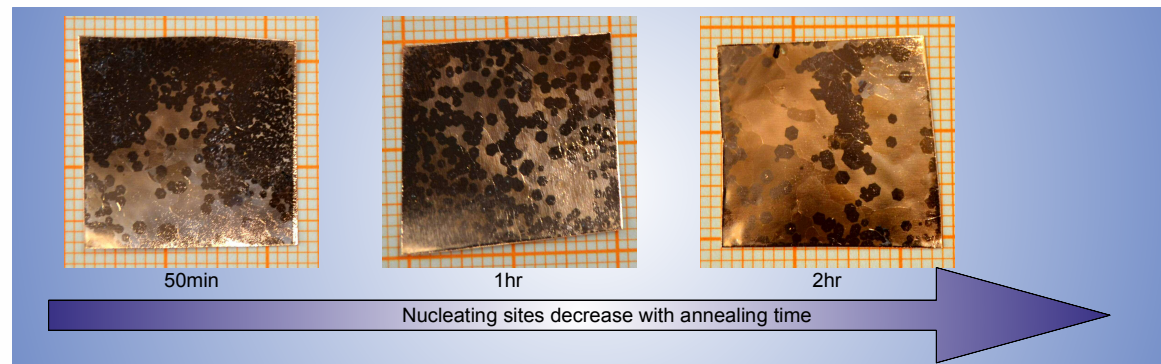
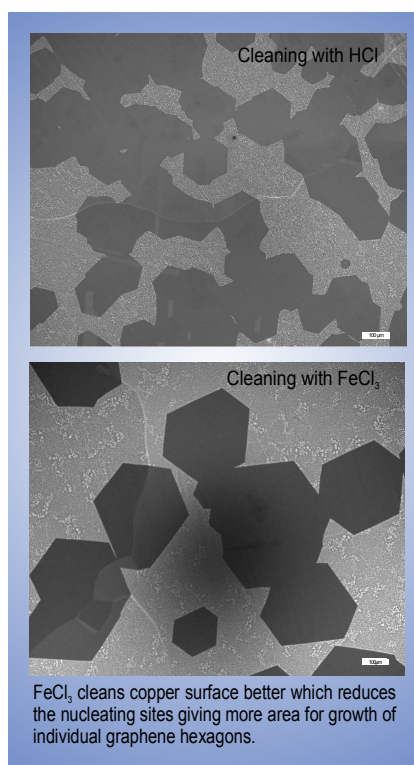
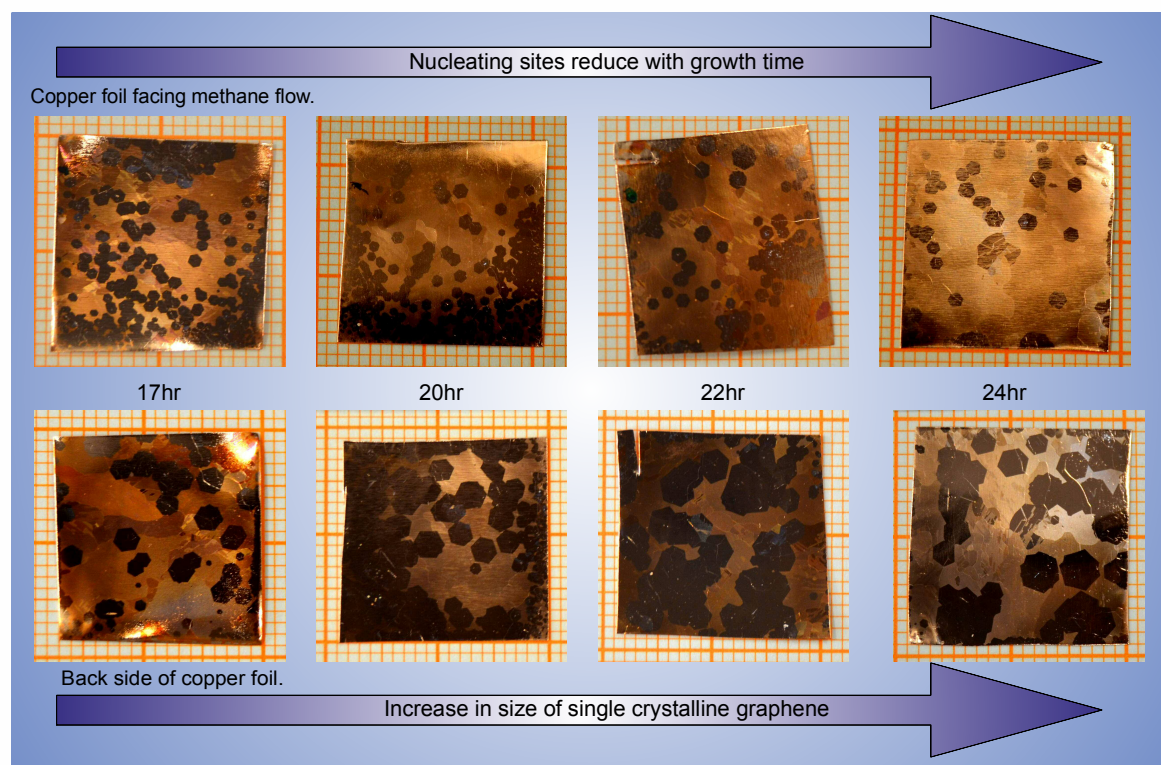


The multiple nucleating sites reduce with oxygen in copper increasing single crystalline graphene.



CVD furnace during growth at 1050°C.

Control Over the Graphene!



Desired quantity and quality of graphene can be obtained by controlling the parameters!

References

- [1] I. Brihuega, P. Mallet, H. González-Herrero, G. Trambly de Laissardière, M. M. Ugeda, L. Magaud, J. M. Gómez-Rodríguez, F. Ynduráin, J.-Y. Veuillen, *Phys. Rev. Lett.* **109**, 196-802 (2012).
- [2] Da Hee Jung, Cheong Kang, Minjung Kim, Hyeonsik Cheong, Hagil Lee, Jin Seok Lee, *J. Phys. Chem. C*, **118**, 3574-3580 (2014).
- [3] Rui He, ting-Fung Chung, Conor Delaney, Courney Keiser, Luis A. Jauregui, Paul M. Shand, C. C. Chancey, Yanan Wang, Jiming Bao, Yong P. Chen, *Nano Lett.*, **13**, 3594-3601, (2013).

Conclusions

- Oxygen rich copper is important for nucleation of graphene on copper foil during growth.
- With increasing growth time, nucleating sites reduce and the size of graphene hexagons increases.
- Annealing time of copper before growth is important for surface rearrangement of the foil which influences the nucleating sites.
- Increasing the concentration of methane in the CVD furnace during growth increases the size of single crystalline graphene areas.
- The decomposition of methane greatly depends on the temperature. At higher temperature more graphene tends to deposit on the copper surface. Also, the rate of growth is faster giving large areas of graphene.
- The cleaning of copper surface is crucial for nucleation of graphene.